Masrur Sarar
ID:- 23201003
CSE340 sec:- 23
Assignment-3

1)

ABB9609
$=(1010\ 1011\ 1011\ 1001\ 0110\ 0000\ 1001)_2$

Bias $= 2^{6-1}-1 = 31$

Actual exponent $= 21-31 = -10$

$(1.1010101110111001011000001001)_2$
$= (1.7241215705 87)_{10}$

Now, $-1.724121570587 \times 2^{-10}$
$= -0.00168$

2)
$(64.2486)_{10} = (1000000.00111)_2$
$= 1.00000000111 \times 2^6$
$= 1.00000 \times 2^6$

$(49.1832)_{10} = (110001.00101)_2$
$= 1.10001 \times 2^5$
Now,
$1.00000 \times 1.10001 = 1.10001$
$2^6 \times 2^5 = 2^{11}$
$\therefore 1.10001 \times 2^{11}$

3)

~~28.48 = (11100.01111)~2~~~

$28.481 \approx (11100.01111)_2 = 1.11000 \times 2^4$

$-4.021 \approx (-100.00000)_2 = 1.00000 \times 2^2$

$28.481 - (-4.021)$

$= 1.11000 \times 2^4 + 1.00000 \times 2^2$

$= 1.11000 \times 2^4 + 0.01000 \times 2^4$

$= 10.00000 \times 2^4$

$= 1.00000 \times 2^5$

$= 32$

4)

```
feq.d t0, f1, f2
beq   t0, x0, jumpNotEqual
jumpEqual:

    jal x0, endCompare

~~jumpEqual~~
jumpNotEqual:

endCompare:
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